Chapter 6:
Writing IS Research Articles

Scientific Writing

- Why publishing is important for scientists
- How scientific publishing works
- Writing a scientific paper
- Reviews and revisions

Publish or Perish

The purpose of an academic is to produce
and
disseminate research results.

Publications advertise your

- **skills**
- **areas of interest**
- **expertise**

Publications allow your work output to be

- **evaluated**
- **compared**

Publish or Perish

- What's in it for the university?
  - Publications advertise the strengths and areas of interest of the university
  - Publications generate direct income from the federal government from publications
  - Publications by research students are an indicator of research training success and help attract students
  - Publications are a key requirement for course accreditations such as EQUIS or AACSB.

  - 'What helps you succeed helps us succeed!'

Scientific publications –why do they matter?

- Papers are the format we choose to make our research outcomes persistent as additions to the body of knowledge.
- Papers are the main proof of valid research activity.
- Papers are the best way for you to market your skills and creativity.
- Papers are the best mechanism for getting feedback about your research from others.
- Papers helps you become a member of a research community and influence it.
- Writing helps clarify your ideas and fit them into the current state of your field.
- Writing keeps your teaching fresh.

You are what you publish!

- Publications are the most important KPIs of an Academic. The number and quality of your publications will determine, at least to some degree, the level of success you will enjoy in your career.

  - **Number** of peer-reviewed publications
    - E.g., US Asst. Prof. minimum 8+ journal articles in 6 years to achieve tenure
    - E.g., Uni Hamburg expects around 4 or so A+ journal articles for a W2 professor; 6 or more for a W3 professor.

  - **Quality** of peer-reviewed publications
    - Citations to your paper: they track which papers reference your paper
      - Raw data
      - H-index
      - G-index
    - The type and quality of the outlet in which your paper is published
      - Journal Impact Factor
      - Journal ratings

A common misconception

- Publish or perish is not about **succeeding**. It is about **surviving**.

- Dissemination is not impact: You do not suceed as a scientist by getting papers published. You succeed by creating impact.

- What is **impact**?
  - Citations?
  - Rankings?
  - Tweets?
  - Changes in the world that go back to your ideas?

Impact Metrics: Citation Data

- Harzing's Publish or Perish
  - Displays citation data for articles, authors and/or outlets
    - Raw data
    - Indices

- For example: H-index
  - A researcher with an index of h has published h papers with at least h citations each.

Metrics: Journal Impact Factor

- The journal impact factor (IF) is a measure of the frequency with which the "average article" in a journal has been cited in a particular year.

A = the number of times articles published in 2019 and 2020 were cited during 2021

B = the total number of 'citable items' published by that journal in 2019 and 2020

2021 impact factor = A / B

Writing Scientific Papers: What do we write about?

- Research!

  - What does that mean?
    - The **process** and, importantly, the **outcomes** of a **scientific** activity.
      - This typically means that we write about research that has been completed.
      - Leads to publication bias: We typically only see “successful research” being published.

    - Very often it is also one of the main ways in which we can paint a picture of ourselves as **scholars**.

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Myth: You can publish anything anywhere

- Publication outlets differ WIDELY in quality
  - Good outlets will only accept good papers describing good research.
  - Publish high-impact papers in high-quality outlets.

- Open publishing process
  - Once you publish, your ideas are out there!
  - Can be good or bad for reputation

- One shot option:
  - Once you published a result, you can’t publish it again!

Publication Outlets

- What different publication **formats** exist?
  - books
  - book chapters
  - journal articles
  - conference papers
  - workshop papers
  - technical reports
  - …

- What are the **differences**?

The most important lessons from this session

- Good paper
  - Purposive
  - Goals clearly defined.
  - Reporting of procedures (including flaws) complete and honest.
  - Objectivity clearly evident.
  - Describe the use of appropriate analytical techniques.
  - Conclusions limited to those clearly justified by the data / facts.

- Good research
  - Purposive
  - Goals clearly defined.
  - Procedure replicable.
  - Objectivity clearly evident.
  - Use appropriate analytical techniques.
  - Conclusions limited to those clearly justified by the data / facts.

Two golden rules of publishing

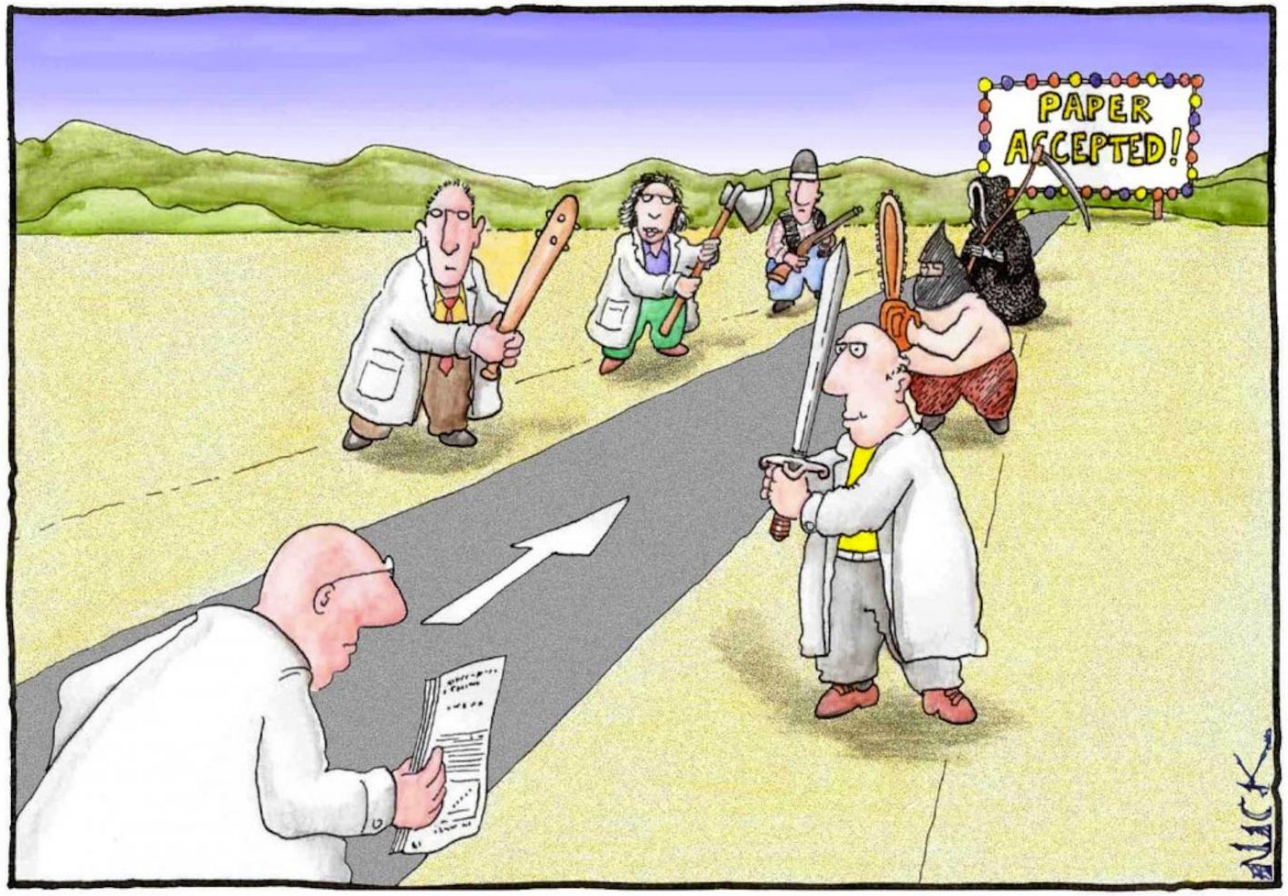
1. You can write good papers only when you do good research.

2. You can waste good research by not writing about it well enough.

What is different in scientific publications from others?

- Scientific publications are peer-reviewed.
  - means writings are subjected to the scrutiny of experts in the same field before a paper will be published.
  - requires a community of experts in a narrowly defined field who are qualified to perform impartial review.
  - Most often double-blind review: authors and reviewers do not know each others' identity.
- The review process is managed by editors.
  - typically a successful senior scientist familiar with the field of the manuscript.
  - Makes decisions about the publication of the manuscript.

Peer review



Most scientists regarded the new streamlined peer-review process as "quite an improvement."

How do we write a paper?

- A scientific paper must contain enough information to enable peers (the scientific community) to:
  - Assess observations/analyses/data/interpretations/conclusions
  - Repeat the experiments/study if they want
  - Evaluate intellectual processes (i.e. are the authors' conclusions and interpretations valid?)

Process for writing a scientific paper

1. Develop an outline with a structure.
2. Revise the structure.
3. Start populating the sections.
   - This typically is not sequential.
   - The first paragraph of the Introduction is the hardest part of a paper to write.
   - Methods is the easiest and can be started even when the research is unfinished.
4. Revise the sections.
5. Revise the paper/thesis.
6. Put it in a drawer, take it out, read again, revise again.
7. Submit.

Why the structure of a paper matters

- Science is about **new ideas in old formats**.
- Reviewers and readers are accustomed to certain ways of reading an article
  - This is called a “script”
- If they encounter a different script, they get scared.
- Thus, in principle, we should follow established scripts and make only mindful variations.

Macro-structure of many empirical papers

Table 1: Overview over Common Headings Associated with Each Section

Section	Associated headings
Theoretical front end	• Background • Conceptual development • Hypotheses development • Literature review • Research hypotheses • Research model • Research questions • Theory • Theoretical background • Theoretical development • Theoretical model
Method	• Analysis • Data collection • Findings • Methodology • Methods • Model testing • Procedure • Research methodology • Results
Implications	• Contributions • Discussion • Future Research • Implications • Implications for future research • Implications for practice • Limitations • Practical implications • Recommendations • Theoretical implications

Tams, Stefan and Grover, Varun (2010) "The Effect of an IS Article's Structure on Its Impact," *Communications of the Association for Information Systems*: Vol. 27, Article 10.

Structuring your Paper

- The Typical Structure of a Research Paper:
  1. Introduction
  2. Theoretical Background/Literature Review
  3. Research Model/Hypotheses/Propositions
  4. Procedure/Approach/Methodology/Research Design
  5. Results
  6. Discussion of Results
  7. Implications – for Research and Practice
  8. Conclusion

Strange, J.R. and Strange, S.M. 1972. "How to read a scientific research report" in Reading for Meaning in College and After, Brooks/Cole: Monterey, pp. 54-66.

Examples

Generic Paper Structure	Example I (Recker et al., 2011) quantitative study published in *MIS Quarterly*	Example II (Seidel et al., 2013) qualitative study published in *MIS Quarterly*
• Introduction	• Introduction	• Introduction
• Theoretical Background/ Literature Review	• Theory	• Theoretical Background
• Research Model/Hypothesis or Theory Development (where appropriate)	• Proposition Development	
• Procedure/Approach/ Methodology/Research Design	• Research Method	• Research Method
• Results/Findings/Outcomes	• Scale Validation • Results	• Interpretation and Analysis of the Case • An Integrated Model of Functional Affordances in the Sustainability Transformation Process • Two Vignettes of Sensemaking and Sustainable Practicing
• Discussion	• Discussion (includes subsections for Limitations and Implications)	• Discussion and Implications
• Limitations and Implications for Research and Practice		• Limitations
• Conclusion	• Conclusion	• Conclusion

Filling the sections: What goes where?

1. Introduction
2. Background
3. Theory/Research Model
4. Method/Procedures/Approach/Methodology
5. Results
6. Discussion
7. Implications – for Research and Practice
8. Conclusion

Section: Introduction

- **Contains three parts, and three parts only:**
  - **Motivation**: What is your problem, and why is it a problem?
  - **Specification**: what is your specific research question/research outcome? What is your research approach (in very general terms)?
  - **Outline**: how will the paper tell us about what you did?

Section: Introduction

- Tips
  - Good introductions are SHORT. The rule of thumb is not more than 1.5 pages with 2.0 line spacing
  - Place the study in the context of previous research but tell only what the reader needs to know to understand the present work.
  - Either avoid jargon or explain it very clearly.

Example

Introduction

The deterioration of the natural environment is a preeminent issue for our society (World Commission on Environment and Development 1987), and business organizations are a main contributor to this challenge (Melville 2010). In responding to increased social, cultural, and also legislative pressures that expand the responsibility of firms to increase attention to environmental concerns (Mintzberg et al. 2002), chief executives have increasingly—in some countries reportedly up to 60 percent (Gadatsch 2011)—committed to "Green IT" initiatives and similar sustainability transformation efforts (Elliot 2011). These initiatives build on information systems as a key resource to assist organizations in transforming to more sustainable entities (Thibodeau 2007), notably because information systems have been argued to be the greatest force for productivity improvement in the last half century (Watson et al. 2010).

Notwithstanding their evident relevance, it remains unclear how exactly information systems can contribute to the sustainability transformation of organizations. Much of the literature has explored specific types of systems and issues related to these systems, including, for example, the role of virtual collaboration systems for green initiatives (Bose and Luo 2011), information systems for managing environmental compliance issues (Butler 2011), or the environmentally conscious design of information systems (Zhang et al. 2011), without examining in detail how information systems can lead to the establishment of environmentally sustainable work practices in an organization.

In this paper, we aim to identify the functional affordances information systems can provide to assist organizations in establishing environmentally sustainable work practices. Functional affordances are potential uses originating in material properties of information systems that identify what individuals might be able to use the system for, given the user's capabilities and goals (Markus and Silver 2008). Identifying functional affordances of information systems that relate to the goal of establishing environmentally sustainable work practices thus assists in understanding how green information systems (Dedrick 2010; Melville 2010; Watson et al. 2010) can be designed that aid tackling sustainability challenges such as energy consumption, waste reduction, resource utilization, or emission management. This knowledge is also important for organizations that need an understanding of how to leverage existing and new information systems in an attempt to better their own environmental footprint. The core guiding research question of our paper is

How do information systems contribute to the implementation of sustainable work practices?

Therefore, the objectives of this paper are (1) to provide an empirical description of the process of developing sustainable work practices on the basis of information systems, and (2) to integrate the empirical findings into a theoretical model grounded in the case data. To achieve these objectives, we conducted a revelatory qualitative case study (Dubé and Paré 2003) with a global software solutions provider that has successfully undergone a sustainability transformation. Based on this case, we develop a theoretical framework that conceptualizes and integrates four key functional affordances that are relevant to implementing environmentally sustainable work practices, namely *reflective disclosure*, *information democratization*, *output management*, and *delocalization*.

Three central contributions are provided. First, we present an empirically developed understanding of the factors pertinent to the implementation of sustainable work practices in an organization, specifically those that relate to functional affordances created through information systems. In the process, we contribute substantively to an understanding of functional affordances created by information systems and how these affordances allow organizations to identify and implement environmentally more sustainable work practices. Second, the paper provides an empirically grounded basis for further theorizing around the transformative power of information systems in creating an environmentally sustainable society. Our argument is that information systems can provide action possibilities that are characterized as environmentally sensible practices. Acting upon these possibilities allows organizations and individuals alike to lower environmental costs and to promote environmentally conscious ways of working. Third, our work generates design advice about the material properties of information systems that are required to provide functional affordances allowing environmentally sustainable work practices.

On the basis of our research, we argue that the primary role of information systems in sustainability transformation is to create action possibilities for sensemaking and sustainable practicing. The creation of these affordances is a key transformative power that information systems provide to organizations in becoming environmentally sustainable.

We proceed as follows. The next section presents a discussion of relevant literature analyzed according to the characteristics of organizational sustainability transformations and the role of information systems within such transformations, and introduces the role of [illegible] and functional affordances as two theoretical lenses relevant to building an understanding of the role of information systems in sustainability transformations. The section that follows details our research method. The next section presents the case interpretation and findings, and the following section

Motivating a Paper: The Gap vs the Hook

- The Gap is usually the argument that something **hasn't been done** yet.
  - This is weak because some things shouldn't be done.
- The Hook is a strategy to find a **problem** that **someone** cares about
  - Can be academic, theoretical, practical…

Grant, A.M., and Pollock, T.G. "Publishing in AMJ—Part 3: Setting the Hook " *Academy of Management Journal (54:5) 2011, pp 873-879.*

Good Gap versus Bad Gap

- Bad Gap:
  - “nobody has studied…”
  - “the literature is silent on…”

- Good Gap = Problem:
  - Resolve a contradiction in the literature
  - Extending the literature to account for specific, important phenomena / understandings / contexts (and why)
  - Solve a puzzle for practice that is important but not addressed by the literature
  - Show how existing literature may mislead our thinking

The Hook Formula for a Clear Introduction

▪ Paragraph 1 – Hook	Motivation, need, ‘hook’
▪ Paragraph 2 – Gap	Situate in research – identify opportunity
▪ Paragraph 3 – Study	How investigate? Process, context & why?
▪ Paragraph 4 – Conclusion	Findings, framework, contribution
▪ Paragraph 5 – Outline with…	Outline the paper: first …, second…, then… and , we conclude

The Formula in 9 Sentences

Hook

1. Sentence stating what the domain is and why it is important.

Gap

2. What is the overall problem or situation in that domain
3. What are the conclusions of existing literature for that problem / situation in this domain?
4. What is the problem or issue with that existing literature?

Study

5. Indicate that this study addresses that problem or issue and state how.
6. Describe the study, sample, and method for addressing that problem or issue.

Conclusion

7. Describe what you found.
8. State explicitly how these findings extend and contribute to existing knowledge.

Outline

9. Describe the overall outline of the paper.

Section: Background

- Provides <u>what is required to facilitate an understanding of your research</u> processes and findings
- Not more. Not less.
  - Relevant theories
  - Relevant concepts/terms that you need
  - Relevant previous methods/algorithms/findings/arguments on which your work is based

Section: Background

- What is it not?
  - A recap or listing of all other works in the area,
  - Every paper that you have read about the area,
  - Criticism of all other works in the area, or
  - A collection of term definitions.

Writing the Background Section

- Tips
  - Organise the section according to topics, not as a list of studies.
  - Discuss related work, rather than just listing it.
  - Explain how your work complements others' work.
  - Explain how your work contradicts previous work.
  - Highlight areas in which your work builds on others' work.
  - Keep it concise.'

  - My rule of thumb: you want to get to your own contributions no later than page 15!

The Background Section - Example (1)

Our research mainly relates to semantic and pragmatic aspects of process models. We complement works that establish a connection between structural metrics and both understanding and error-probability [CGP+05, MRC07, MVD+08, GPP08] by investigating whether the verb-object labeling style might be superior to other styles. **In contrast to previous works like [Mil61, SM01, MCH03]** that recommend verb-object labels, we provide a sound empirical justification for this guideline.

The Background Section - Example (2)

Recent research has started to examine process model understandability, for instance, the impact of process model structure, model user competency, and activity node labeling. While the impact of structural properties is clearly identified [4], it is also reported that model readers systematically overestimate their ability to draw correct conclusions from a model [3]. Furthermore, shorter activity labels have been found to be positively correlated with understanding [5]. **This raises the question in how far** a better representation of domain semantics in process models beyond the use of text labels only would improve understanding.

In our work we continue along this line of work towards more understandable process models. Current practice indicates that the labeling of activities is a rather arbitrary task in modeling initiatives and one that is sometimes done without a great deal of thought [8]. This can undermine the understandability of the resulting models in cases where the meaning of the labels is unclear, not readily understandable or simply counter-intuitive to the reader.

The Research Model/Theory/Hypotheses Development Section

- This is where the magic happens but it’s not part of every paper.
- This is NOT where you discuss an existing theory but where you develop NEW theory.
- Introduces, arranges and argues an overview of the factors being studied in an empirical setting
  - The independent and dependent factors in an experiment,
  - The important theoretical constructs and their relationships in a field study, or
  - The set of hypotheses or propositions to be tested/explored/falsified

Section: Research Model

How do we organize the research model section?

- Start with an overview of conceptual or research model being developed
- Discuss components of the model in paragraphs with relevant subheadings that reflect the model
- follow a disciplined structure that moves from concepts to associations to laws to boundaries
- Each hypothesis/proposition must be preceded by a strong argument as to why it should hold.

Tips for Theory/Model Development

- Explain concepts or constructs in **tables**

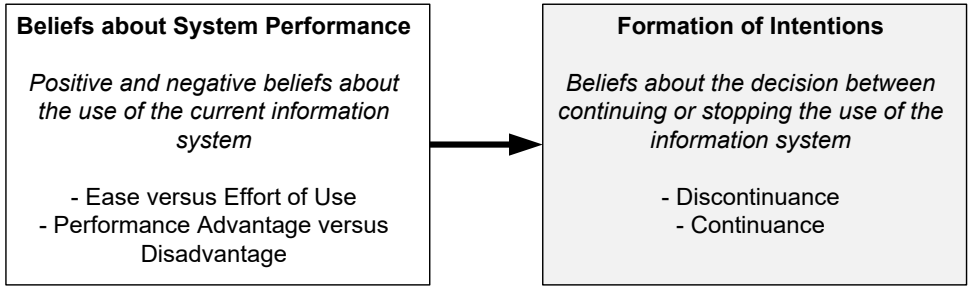
Table 1: Conceptualization of *Green IS practices*.

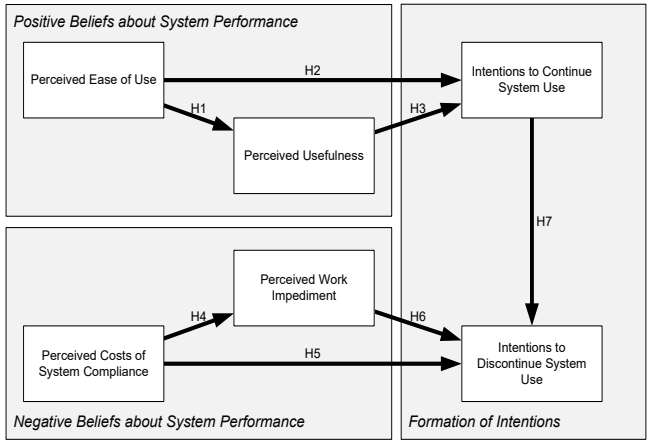
Second-order construct	First-order constructs	Definition	Description
Green IS practices	*Process re-engineering*	IS-enabled reengi-neering of business and production processes	Green IS practices that enhance the resource efficiency of business and production processes through IS-enabled process re-engineering and business transformation.
	Environ-mental management systems	Use of IS-based environmental management systems to control resource flows, waste, and emissions	Use of IS-based environmental management systems that track resource flows, waste, and emissions (to provide information for environmental control and sustainability-oriented decision-making); enhance transparency; and provide aggregated information for external stakeholders through environmental reports.
	Environ-mental technologies	IS-enabled environmental technologies that reduce the footprints of products and services	Improvement of the environmental characteristics of end products and services with the help of Green IS, such as smart buildings, traffic management systems, smart grids, engine control units, and dematerialization through digital services.

Loeser, F., Recker, J., vom Brocke, J., Molla, A., and Zarnekow, R. 2017. "How It Executives Create Organizational Benefits by Translating Environmental Strategies into Green Is Initiatives," *Information Systems Journal (27:4), pp. 503-553.*

Tips for Theory/Model Development

- Differentiate between conceptual level and empirical level





Recker, J. "Reasoning about Discontinuance of Information System Use," *Journal of Information Technology Theory and Application (17:2) 2016, pp 101-126.*

Developing Hypotheses

- A hypothesis must contain a justified theoretical argument for why you expect a certain phenomenon to occur (or not)
  - No such thing as a self-evident hypothesis
- Connect your hypotheses to existing research
  - Include references to key literature in building your argument -- this literature should have been discussed in your background section
- Keep your hypotheses simple but precise
  - They should specify two (or more) key constructs and the relationship between them.

Example: Developing a hypothesis

Example: Hypothesis development

Explanation of underlying mechanism

Evidence supporting the mechanism explanation

Specification of variables and their relationships

Justificatory mechanism

Model interpretation occurs in two stages (Newell & Simon, 1972), these being perceptual processing (seeing) and cognitive processing (understanding). During perceptual processing, computational offloading effects can occur that can aid the subsequent cognitive processing. Model elements that can be perceptually discriminated reduce the processing burden of the cognitive system because the parsing of model elements to different semantic components is performed by the perceptual sensors (Treisman & Gelade, 1980). **Several studies suggest that perceptual processing largely explains differences in the effectiveness of the subsequent cognitive processes** (Larkin & Simon, 1987; Petre, 1995).

These considerations suggest that **the perceptual discriminability effect of an explicit visual syntax** of Gateway constructs **will be stronger** when **model complexity** is increased. […] When perceptual discrimination of all the constructs in a complex process model is not aided by the visual syntax (as in the case of an implicit representation for control flow divergence and convergence, **cognitive overload ensues due to the limited capacity of the working memory, leading to rapidly decreasing comprehension** (Miller, 1956).

Recker, J. (2013). Empirical Investigation of the Usefulness of Gateway Constructs in Process Models. *European Journal of Information Systems, 22(6), 673-689.*

Developing Hypotheses: Tips

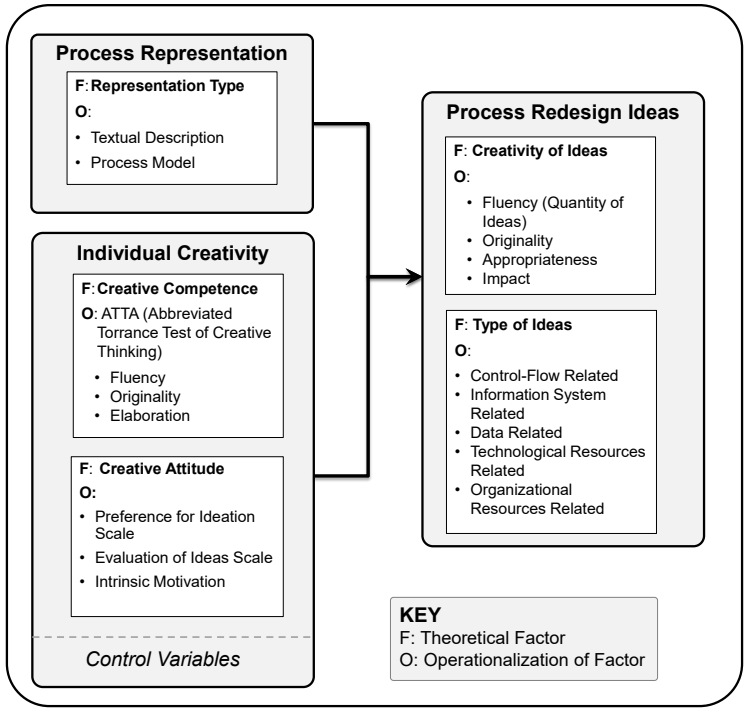
- Do not use words like "prove”. Instead use words like “suggest” or “support”.
  - Hypotheses are never proven; only supported.
- Explain your expectations clearly.
  - Identify dependent and independent variables and the direction of expected relationships.
- Make sure the **relationships between the variables** are testable.

Example: Presenting a research framework

The model shown in Figure 1 frames our primary research interest: the influence of the type of process representation on the *creativity* and *type* of the process-redesign solutions. Based on findings in the literature on how individual characteristics relate to creative problem-solving processes, the model acknowledges the relevance of the individual as a creative person by using creative competence [16] and creative attitude [4] as control variables.

…

Based on our model, we present two sets of hypotheses that describe our expectations about the effects of a type of representation in a process-redesign task on the solutions conceived in this task. First, we explore whether the quantity and quality of process-redesign solutions varies.



Figl, K., and Recker, J. 2016. "Process Innovation as Creative Problem-Solving: An Experimental Study of Textual Descriptions and Diagrams," *Information & Management (53:6), pp. 767-786.*

Section: Method/Approach/Research Design

- In this section, you describe how your research was carried out.
- Describe it in such a way that another research can repeat your work based on your description (to meet the requirement of replicability).
- This section should follow a well-established script and is relatively easy to write.
  - Experiment paper: experimental design, variables and measures, materials, tasks, procedures, and participants.
  - Survey paper: design, measures, sampling strategy, procedures, and validation of measures.
  - Qualitative paper: empirical setting, modes of data collection, and processes and techniques employed in data analysis.
  - Design science papers: overview, meta-requirements or design principles, artefact description, and evaluation.

Section: Method/Approach/Research Design

- Typical key elements
  - Research strategy
  - Research materials
  - Case sites
  - Scope of data collection
  - Sampling strategy
  - Participant selection
  - Measurement strategy
  - Analysis strategy

Section: Results/Findings

- **Only contains a description of findings**.
- Should contain just factual result reporting (in the past tense) including statistical (or other) tests, but no discussion.
- Writing, statistics and graphs should be presented as simply as possible.
  - Try to make figure legends self-contained so that reference to the text is unnecessary.
- Do not present irrelevant data to which the Discussion will not refer and vice versa.

This section is easy to write, and quite methodical – check published papers that use a method similar to yours.

Section: Discussion

- When most people read a paper, they read the Abstract first, then the Introduction, some graphs or tables and then the Discussion.
- Therefore, the Discussion should begin by summarising the main findings.
- Then: we interpret the findings
  - This is about **sensemaking** of the findings or results: What do they actually mean? Why did you get the results that you obtained?
    - **Explaining** the results: why did you find what you found?
    - **Abstracting** the results to concepts: what does it mean in a larger view?
    - **Theorizing** the results: what do the results tell us about the existing/new theory?
- Keep the discussion to the results, i.e., **do not go beyond the data.**
- Use present tense.

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Examples – Summarizing the key findings

This study provides empirical results on the associations that two types of characteristics of the BPMN modeling grammar, *i.e.*, perceptions of a lack of ontological completeness and ontological clarity, have with user beliefs about the usefulness and ease of use of the modeling grammar.

Discussion

The objective of this study was to examine the utility of an extended model of continued process modeling grammar usage behavior. Data collected from an online survey of 529 current BPMN process modeling grammar users were used to test the model. The theoretical model demonstrated adequate fit with the data. Most causal relationships in the model were found to be significant as hypothesized. We identify a number of interesting results.

Examples – Interpreting the key findings

While we found that users who did not use constructs that are classified as excess had higher perceptions of the ease of use of the grammar, **the differences in PEOU ratings between the groups were not always significant**.
One possible interpretation of these results is that some of the excess constructs in BPMN (e.g., Association Flows or Text Annotations) are used by modelers to mitigate or mask other deficiencies (e.g., construct deficit or overload). These workarounds may distort the theoretically hypothesized results, in that users may have positive usage perceptions about some of the excess constructs. For example, a modeler may find the excess construct Text Annotation useful because it allows them to handle other deficiencies of the grammar (for instance, those relating to the deficit for articulating business rules).

Overall, these findings attest to the importance of adequate training in process modeling. Training serves to reduce uncertainty about a grammar by providing information about the features, nature and characteristics of the grammar. Greater learning thereby can amplify perceptions about the usage of a grammar in a positive direction. Also, greater learning can establish self-efficacy beliefs in the users, which also helps rectifying potential problems in the use of the grammar (as shown by the moderating effect of grammar familiarity). Our study suggests that it could be possible for organizations and individuals alike to increase user abilities in process modeling with the view to establishing positive usage beliefs.

We further found that, when model complexity is increased, syntactic effects related to the perceptual discriminability of the Gateway constructs have a positive effect on model interpretability, over and above the effects of the constructs' ontological deficiencies on a semantic level. Given that 'typical' process models found in industry practice are quite complex and may involve up to thousands of activities and related objects such as data and applications (Mendling, Reijers, & van der Aalst, 2010), this finding is significant in that it underlines the importance of perceptual, syntactic considerations in the design of process models that are readily and intuitively perceptible by the intended audiences. **Our results suggest that potential semantic deficiencies of these models (in terms of the precision of the clarity of their specification) can be positively mitigated through appropriate visual means, e.g., by selecting easily distinguishable shapes for better perceptual discrimination.**

Section: Implications

- This section is absolutely important. It should be explicit in the structure.
- Good journals <u>demand</u> specific implications for ongoing research and practice.
- Implications for Practice
  - Detail how the study findings **change/impact** the way that important **stakeholders** work in **actual practice**.
- Implications for Research:
  - What have we found that can guide future research
    - E.g., a new measurement instrument
    - E.g., a new theory
    - Potential extensions of the study
  - What are future research opportunities that continue from the study described
    - New questions that arise from the way the research question was answered in the paper.

Example

Example: Implications for Research: Future Research Opportunities

Different independent variables

Different measurements

We chose one semantic characteristic, construct excess as a manifestation of ontological non-clarity, and one syntactic characteristic, perceptual discriminability. **Thus, opportunities exist for fellow scholars to examine different semantic and syntactic characteristics of process models**, and the impact these characteristics may have on the ability of individuals to understand the models.
Other research could extend our approach to measuring process model interpretability. […] We focused on individuals' understanding of grammatical elements and their meaning in a process model (surface understanding), which is fundamental to being able to faithfully and efficiently interpret a process model. **Future work could now extend this work and examine the problem solving performance** (which is indicative of deep understanding, see Burton-Jones & Meso, 2008; Gemino & Wand, 2005) of individuals […].

Example: Implications for Research: Changing the Conduct of Research

New methods

New outcomes

First, our framework suggests **a reinvigorated focus on CM design research**. The framework centers on the mediating role of CM scripts to facilitate transitions between states of reality. […] Traditional CM grammars were designed with a focus on representing the transition from physical reality to digital reality alone. To extend the mediating role of CM scripts (e.g., from digital to digital), researchers may need to develop new grammars and methods, such as new grammars with new constructs to present states in digital reality, and methods for tracking changes in the states between realities.

A second broad implication of our framework for CM scholarship research concerns its **call for research on dependent variables that are new to CM**. The key dependent variables of quality and understandability were traditionally evaluated based on the scripts' underlying grammars. Future research should evaluate new dependent variables like the quality, consistency, traceability, and understandability of newly developed scripts and grammars.

Example: Implications for Practice

Target audience

What they should change

Furthermore, **providers of training** and **analysts in charge of conventions** can be informed through our study about the role of gateway constructs in process modeling. Our results suggest that gateway constructs are helpful in aiding process model interpretation in cases where the depicted processes are quite complex but not in cases where the processes are straight-forward. **Modeling conventions could leverage these insights by developing a staged approach for process modeling grammar usage dependent on process complexity**. For instance, managers in charge of modeling conventions can use the findings of our study to define a restricted set of process modeling grammar constructs to be used for complex processes, so as to improve the level of understanding of the final process models.

Section: Conclusion

- Provides the closing frame to a paper.
- Is often little more than one abstract statement.
- Recaps very briefly the key contributions of the paper, without going into detail.
- Reflects about the research presented in the paper.
- Never introduces new material or findings.
- Rule of thumb: don’t cite anything!

Example

Example: Conclusions

Brief summary of contributions

Recap of findings

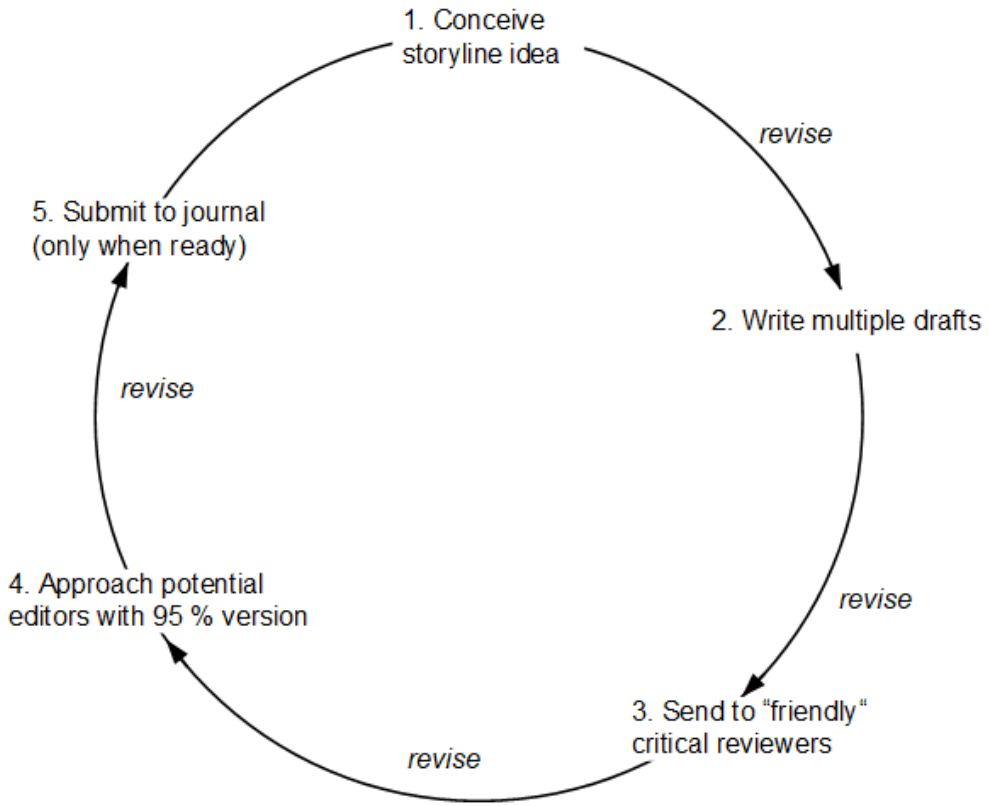
Reflection

In this study, we contribute to process modelling research by providing a theoretical and empirical analysis of the ontological deficiencies of the BPMN grammar. **Our study provides theoretical and empirical evidence** on the strengths and weaknesses of using BPMN, and it also discusses a range of contextual factors that impact the use of BPMN.

These **findings assist in understanding** the capabilities and issues in the use of BPMN and can guide executive decision makers in their investment decisions when they implement a process modelling initiative. In summation, our study presents a rich and contextualised explanation of the usage behaviour that is associated with process modelling grammars and provides further evidence of the utility of the ontological theory of grammar expressiveness in aiding our understanding of process modelling practice.

Wrap-Up: The Article Writing Life-Cycle

- Conceive idea
  - Revise
- Write multiple drafts
  - revise
- Send to “friendly” critical reviewers
  - revise
- Approach potential editors with 95% draft
  - revise
- Submit
  - only when ready



What happens when you submit your manuscript?

- Submission is commonly by electronic upload. Most journals use ManuscriptCentral; or EES or something similar.
- A journal editor will determine if your manuscript content is potentially suitable for publication in the journal.
- If it is, the manuscript will generally be sent to senior/associate editors and/or two or more reviewers.

Why Editors Desk-Reject Papers

- Lack of fit to the journal
  - Doesn’t cite ANY papers in the journal
  - If substantive journal: paper is about a different domain
  - If general-level: paper is not general enough

- Important: lack of fit is not an evaluation of quality!

Why Editors Desk-Reject Papers

- Lack of Likelihood
  - Will the paper have a good chance of successfully traversing the review process?
  - What are the odds that the paper will receive only rejects?
  - Protection of scarce resources: the reviewers!

Why Editors Desk-Reject Papers

- Lack of Formalities
  - Do papers adhere to the submission requirements?
  - Depends on journals: some are VERY strict.
  - Is the language good enough?
  - Don’t take this lightly.

If not: wait… for the editor's decision

- **1. Rejection** – the paper is not accepted by the journal – the editor will tell you why. This decision is final.

- **2. Request to revise and resubmit** - The manuscript is denied publication in its current state, but the editors and reviewers feel that a reworked manuscript *may possibly* be accepted if submitted to the journal again for new evaluation, typically by the same editor and reviewers.
- - often distinguishs between *major or minor revisions*

- **3. Acceptance or conditional acceptance -** the editor has decided that the paper can be published as is, or with changes so minor that it does not require further review.
- - Usually only about 5-10% of submissions get accepted by (good) journals.

Understanding Reviews

MIS Quarterly

Evaluation Form

Reviewer: [insert your name here (voluntary)]

Manuscript Name: [insert title of paper here]

Part I.

Using the "Standards for Reviewing" on page 2, please express your level of agreement with each of the following statements, indicating your choices with an "X":

Criterion	Strongly agree	Agree	Slightly agree	Unsure	Slightly disagree	Disagree	Strongly disagree
1. The paper is conceptually significant.							
2. The paper is practically significant.							
3. The paper is well designed.							
4. The design is well executed.							
5. The presentation of ideas is effective.							
6. Overall, the paper presently makes a major contribution.							
7. Overall, the paper has the potential to make a major contribution.							

Overall Recommendation

Please make a recommendation about the disposition of the manuscript ranging from (1) "reject," (2) "reject but invite new submission" (i.e., very risky revisions), (3) "major revisions," (4) "minor revisions," (5) "accept conditionally" (i.e., very minor revisions), and (6) "accept."

	Reject	Reject and resubmit	Major revisions	Minor revisions	Accept conditionally	Accept
Overall evaluation.						

Page Break

Standards for Reviewing

1. **Conceptual Significance:** The work represents an important contribution to knowledge. It extends or challenges IS theory, empirical literature, methods, IS professional issues, or IS body of knowledge. Ties to relevant literature are clear as is the thrust of the central argument. The work explicates underlying assumptions well and provides direction for extending or improving on the present work.

2. **Practical Significance:** The work contributes to our understanding of current technological and organizational problems or challenges faced by IS or other practitioners.

3. **Design and Execution:** Methods, subjects, logic, and techniques (where relevant) are well designed for the investigation of the questions posed. The work is well executed, including provision of pertinent evidence and interpretation of results. Where appropriate, operationalizations of theoretical constructs, validity, and the choice of statistical and/or mathematical analysis are well done. The work adheres to AIS and generally accepted codes of scientific ethics.

4. **Presentation:** The work adopts a professional style and tone and is concise. It is grammatically correct and clear in its use of figures and tables. The flow of ideas in the paper is logical and there is a clear tie between its use of prior literature and a clear link between the method it adopts and its conclusions. The work is presented at a level of sophistication and length appropriate to the readership of the journal.

Page Break

Revising a paper based on reviews.

- Don’t be precious – get published.

- If the paper is **rejected** – determine why.
- If there are legitimate flaws – fix them.
- Do not resubmit the same flawed manuscript somewhere else.

- If you are asked to **revise and resubmit** – do it.
- Take all editorial and reviewer comments seriously.

Revision Management

1. Put the reviews aside for a while.
   - All initial reviews are critical
     (you will very rarely accepted straightaway – it only happened to me once)
   - You will probably have some strong negative reactions to a review.
   - After several days/week, they will look more manageable.

2. Read the reviews.
   - You need to truly and fully understand each and every comment.
   - Read the reviews several times.
   - Develop a strategy

3. Tabulate reviews.
   - Create a table for the editors and reviewers and copy and paste each “block” of review comments into a separate row.
   - Have three columns, one for the comment, one for the response, and one for notes and discussion within the research team.
   - Mark them as “quick fixes” (easy to handle) and “major pieces of work”.
   - Indicate the type of revision required: (a) change the research, (b) change the presentation of research, or (c) a suggestion that cannot be handled given the research conducted.

Revision Management

4. Revise the paper.
   - Changes will have to be made to the document.
   - Changes will be more, or less, ranging from changing/inserting/deleting paragraphs/figures/tables of some sections, to writing a completely new manuscript.
   - Don’t be afraid to make big changes to a manuscript. A major revision should always look like one.
   - Some suggest completely re-writing a paper from scratch for major revisions – and in my experience this can often be necessary.

5. Write a response letter.
   - Write point-by-point responses to all comments made.
   - Write the response letter by perusing the revision table.
   - Include a pre-amble that expresses your gratitude towards the editor and reviewers for considering your manuscript and investing their time to give you and assisting you to improve your work.
   - Outline the strategy your pursued in your revision.
   - Summarize the main changes made.

Examples for Response Letters
1) Expansive

	to the emergence of this affordance. Please see the revised discussion on pages 29-32.
Conceptualization of Delocalization Affordances is unnecessarily constrained – Delocalization affordance are those that "enable work practices to become location independent". This is more bounded than the other three types. Why? For example, it is argued that delocalization affordances allow employees to rethink work practices. What is special about location-related	The argument here is essentially one of how much our concepts abstract from the evidence collected. We concur that delocalization on a broader level could be viewed as a form of expansion or enablement. In such case, we agree, such affordances might also relate to work changes that reduce resource consumption in terms of staff or other resources. We thus followed this reviewer's recommendation in that we explain why location-relatedness, indeed, is critical in the context of sustainability transformations. This information is now found on page 32. Still, we would argue against a more abstract conceptualization on two counts: First, our evidence strongly suggests the prevalence of technology-enabled changes that

- Interprets review comment
- Outlines revision alternatives
- Justifies chosen revision
- Describes revision.

Examples for Response Strategies
2) Concise

- Is direct and blunt.
- Only states what has been/not been done.
- Doesn’t go into detail.
- Offers brief justification.

Comment:
Do program chairs adjust for the tracks that accept too few or too many papers?

Response:
We **do not have concrete data** to answer this issue. We believe based on our own experiences in conference chairing and reviewing that this is often the case. However, we do not have data on this **and therefore we have made no changes** to the paper based on this comment.

Example: Revision Cover Letter

Response to the Comments by the Editors
Submission to the [JOURNAL], [SubmissionID]

[Paper Title]

Dear Editorial Team

Many thanks for providing us with constructive and comprehensive feedback about our submission. We found the responses very valuable in further improving our theoretical model and data analysis, as well as in revising the way we describe our research in the manuscript. **We detail our responses in the tables below**.
In summary, our main changes are as follows:

1. We expand on the conceptual foundation for our work and provide more precise and explicit construct definitions.
2. We expanded on the discussion of findings, with a particular view on improving on managerial implications and design advice that follows from our work.
3. We tested alternative conceptualizations and measurement models to examine the suggestions by the reviewer and to attest to the validity of our proposed model.
4. We revised the structure and content of our manuscript and streamlined the flow of the paper by moving details about our analyses to a separate appendix.

We are confident that these changes significantly improved the paper and **we hope that our efforts become evident** from the responses below as well as the revised manuscript itself.
We look forward to receiving your feedback.

Best regards,
[the authors]

End of Chapter 6

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